

Lecture 4

–

Missing Values

Haussperling (Spatz) / House Sparrow

(*Passer domesticus*)

- It is a small bird that has a typical length of 16 cm. Females and young birds are coloured pale brown and grey, and males have brighter black, white, and brown markings.
- The house sparrow is strongly associated with human habitation and can live in urban or rural settings.
- Because of its numbers, ubiquity, and association with human settlements, the house sparrow is culturally prominent.
- Most house sparrow vocalisations are variations on its short and frequent chirping call. Transcribed as *chirrup*, *tschilp*, or *philip*, this note is made as a contact call by flocking or resting birds; or by males to proclaim nest ownership and invite pairing.

Sources:
* Photo by Susanne Jutzeler on pixabay
* https://en.wikipedia.org/wiki/House_sparrow

Focusing Objectives

By the end of today's lecture, you will be able to:

1. Understand different missing data mechanisms and their impact
2. Detect missing values using Pandas functions
3. Apply various techniques to handle missing data effectively
4. Understand the impact of different handling strategies on dataset quality



Missing Values

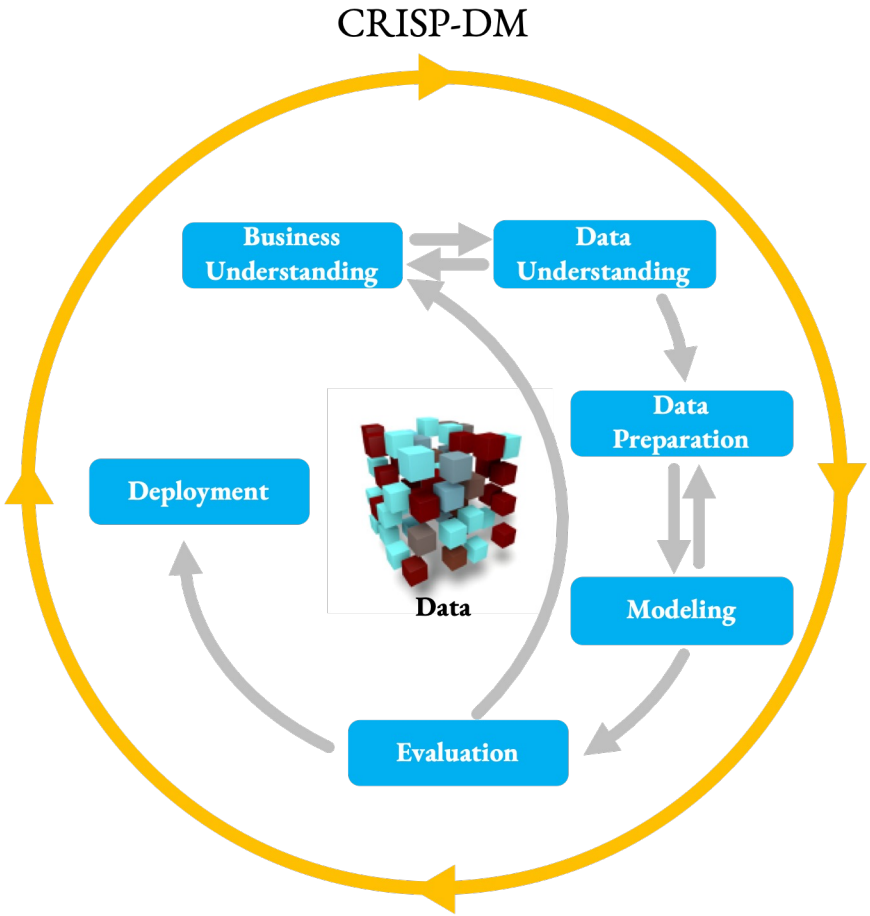
1. CRISP-DM: Data Understanding and Preparation – Missing Values
2. Reasons for and Consequences of Missing Values
3. Detecting Missing Values
4. Types of Missing Values
5. Handling Missing Values

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Data Understanding and Preparation Phases of CRISP-DM

Data Understanding & Preparation

- Main goal: gain a comprehensive understanding of the data that will be used for the data mining project and prepare the data for modeling by cleaning, transforming, and selecting the appropriate variables
- Data quality is being analyzed and improved (using data cleaning methods)
- Additional attributed are created as needed



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Missing Values

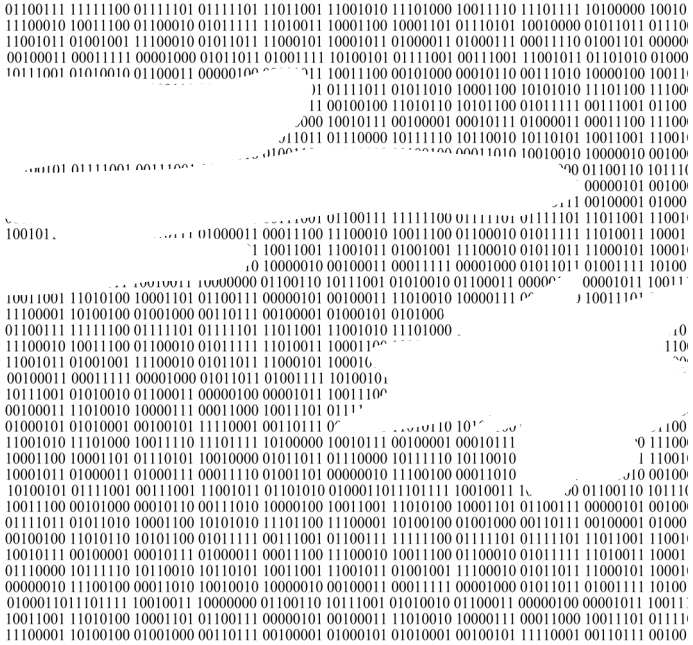
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Motivation – Missing Values

Missing values

- Are **very** common in real-world dataset
- Are very often are not treated properly
- Can cause severe problems when not treated properly
  - Can adversely affects model performance
  - Can lead to bias if the missing values are not random
  - Can reduce statistical power, leading to misleading conclusions
  - Many ML algorithms cannot handle missing values (e.g., k-NN, SVM, Neural Networks)



Sources of Missing Values

- Real-World Examples of Missing Data
  - **Medical Records**: Some patients may not report certain symptoms
  - **Financial Data**: Transaction data may have gaps due to errors or privacy concerns
  - **Survey Data**: Respondents may skip questions they don't want to answer
  - **IoT and Sensor Data**: Sensors in weather stations, industrial machines, or smart homes may fail to record data due to connectivity issues, battery depletion, or hardware/software malfunctions
  - **E-Commerce and Customer Data**: Online retail platforms may have incomplete customer profiles if users do not provide optional details (e.g., phone numbers)

➔ Typical reasons for / sources of missing data
  - Something crashed and data was not recorded for a certain amount of time
  - Users chose not to fill out a field
  - Data was lost or corrupted
  - Bug in the code

➔ Some of these sources are just random mistakes

➔ Others caused by a deeper, underlying reason

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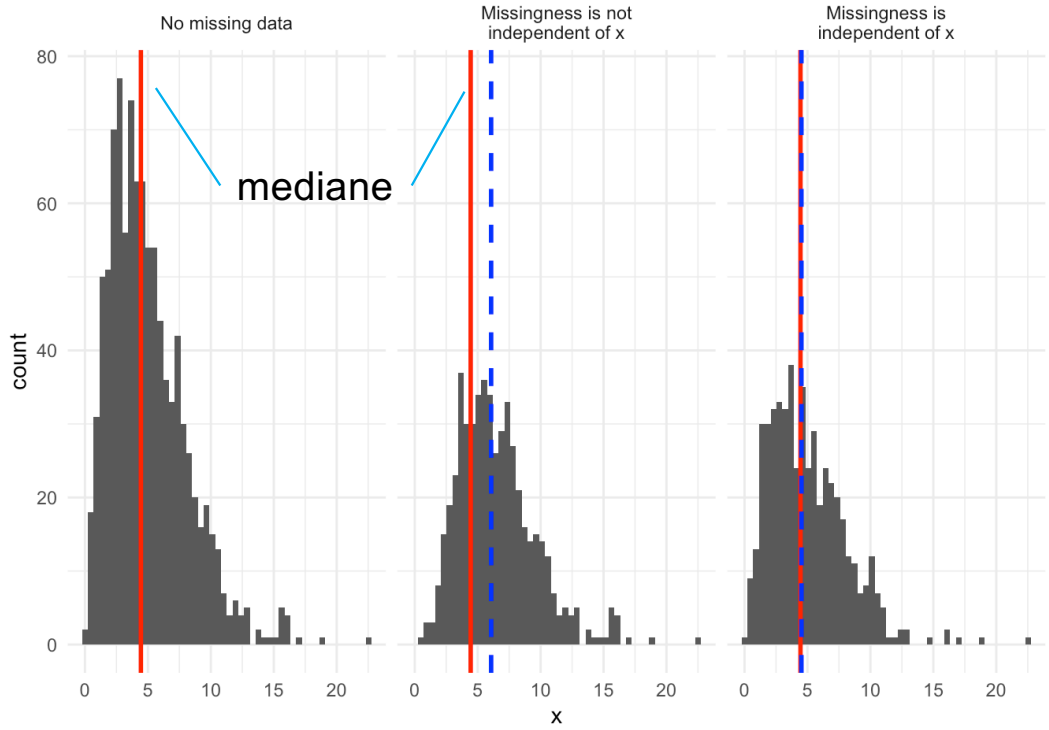
Consequences of Missing Values → Bias

- Missing data can induce bias in the estimates of descriptive patterns and causal effects!

Example

- Describe income distribution in a country with survey data
- Some individuals' incomes are missing
- Suppose low-income individuals are less likely to report their income than high-income individuals
  → thus, missingness concentrates in the lower portion of the distribution
  → characterization of the income distribution biased!

Summary: Missingness correlated with the variable we are trying to describe → characterization of the median of the distribution is biased



Source: https://egap.org/resource/10-things-to-know-about-missing-data/

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How to identify missing values

1) NaN Values (Not a Number)
   - pandas encodes missing values as np.nan (or short, NaN)
   - pandas provides many methods to identify and handle NaN values

2) Placeholder Values
   - Invalid Values (often 0 or negative or out-of-reasonable-range like age 999)
   - Special Values (PLZ 12345, 11111, 99999 or 27498: “Handelsmärkte für Helgoland”)

lurking variable here could be cashier who has recorded the value, e.g. one cashier uses 11111 or 99999 as out of range value

➔ Domain knowledge critical

NaN Values: Pandas functions to identify NaN values

- `df.describe()` → Helps detect NaNs in numerical columns
  - `df.describe().T` → transposes the output, easier to read

- `df.isna()` → Returns a boolean mask
  - `df.isna().sum()` → Counts missing values per column
  - `df.isnull()` is an alias for `df.isna()`

- `df.info()` → Shows non-null counts (=non-na counts) for all columns

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Placeholder Values: recommended approach using pandas functions

1) Analyse value distribution to identify placeholder values:
   - `series.value_counts()` → Return a Series containing counts of unique values
   - Sorted in descending order of frequency (use `sort_index()` to sort by value)
   - For continuous attributes, use the `bins` parameter to reduce the number of values
     Example: `df.attr.value_counts(bins=20).sort_index()`

2) Check, how often a potential placeholder occurs:
   - `(series==value).sum()` → Returns how often `value` occurs
     Example: `(df.attr==0).sum()`

3) Replace placeholder value by NaN (storing the result into a new dataframe)
   - Example:
```
clean = df.copy()
clean[['a1', 'a2']] = clean[['a1', 'a2']].replace(0, np.nan)
```

Exercise

Exercise 1

Pima Indian Diabetes Dataset – Detect Missing Values

check data with describe to see min/max > plausible ranges?
check values individually if suspicious min/max and gaussian
normal distribution (if recognizable), else if plausible value ranges
replace impossible 0 values

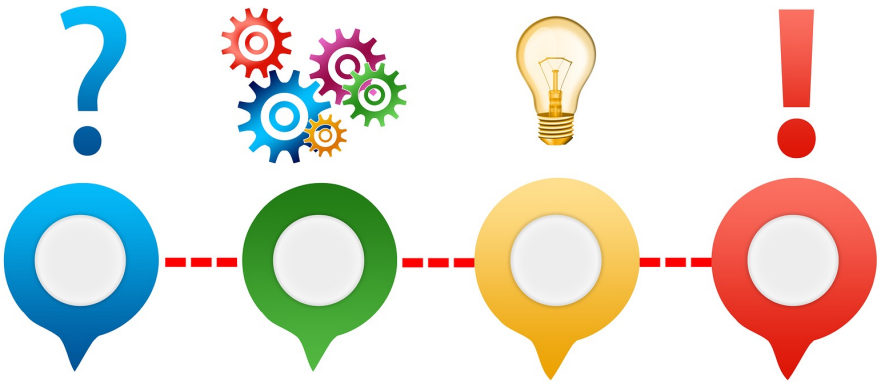


Photo by Gerd Altmann on Pixabay

Dataset: Pima Indians Diabetes Database

- From the National Institute of Diabetes and Digestive and Kidney Diseases.
- Business Problem: diagnosing diabetes
- Data Science Problem:
  - Idea: predict diabetes based on easy-to-measure tests
  - Fundamental task: Classification
  - Data:
    - Several predictor variables e.g., the number of pregnancies the patient has had, their BMI, insulin level, age, ….
    - Target variable: Outcome.
    - Constraints on the selection of the patients (taken from a larger database), e.g., all patients females at least 21 years old of Pima Indian heritage.

Number of Instances: 768

Number of Attributes: 8 plus class

Attributes:

1. Number of times pregnant
2. Plasma glucose concentration a 2 hours in an oral glucose tolerance test
3. Diastolic blood pressure (mm Hg)
4. Triceps skin fold thickness (mm)
5. 2-Hour serum insulin (mu U/ml)
6. Body mass index (weight in kg/(height in m)^2)
7. Diabetes pedigree function (scores the likelihood of diabetes based on family history)
8. Age (years)

9. Class variable (0 or 1)

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Types of Missingness

MCAR:
Missing Completely At Random

Assumption: fact that a certain value is missing has nothing to do with its hypothetical value and with the values of other variables.
In practice, this is assumption very rarely holds.

Example:
- in a medical study, a student drops a random blood sample

MAR:
Missing At Random

Probability that a value is missing depends on the set of **observed features** but is neither related to the specific missing values nor to other **not observed features**.
This is the most common assumption used in practice.

Example:
- in a survey, women more often refuse to give their age then men (and the gender is recorded)

MNAR:
Missing Not At Random

The missingness of a value either depends on the **missing value itself** or on **unobserved features**, which is problematic.

Examples:
- in a survey, people with very low income leave the income field empty more often, then people with higher incomes (missingness of the income depends on the missing income value itself)
- in a survey, women more often refuse to give their age then men (and the gender is not recorded) (missingness of the age depends on the gender)

Source: Donald B. Rubin, Inference and Missing Data; Biometrika (1976), 63, 3, pp. 581-92

1. hängt vom attribut selber ab
2. hängt von einem anderen (nicht aufgenommenem) attribut ab

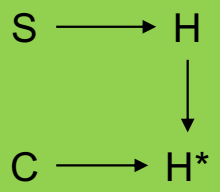
lurking variables

Categories of Missing Values

independent of attributes

MCAR: Professors cat chews on random homeworks

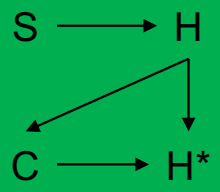




dependent on another attribute that is recorded

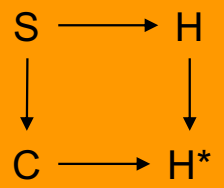
MAR: Professors cat chews on homeworks more than 2 pages long





dependent on attribute that was not recorded

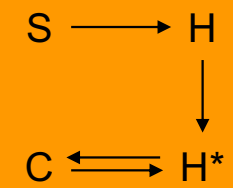




MNAR (on not collected features): Professors cat (very sensitive) chews on smoking students homeworks

dependent on attribute itself





MNAR (on missing value itself): Professors cat (very smart) chews on bad homeworks

S, H, C, H*: Sets of Attributes (S of the Student, H of the complete Homework, C of the Cat, H* of the Homework with missing values), Arrows: Dependencies

Deciding on the Types of Missingness

- Primary source: domain knowledge!
  - How has the data been collected?

    where does it come from?
    how was it accumulated?
    reason for format, origin, etc

    analysis of missing value tuples could reveal correlation to how the samples were collected, e.g. taking blood pressure and skin thickness may need training, if both are missing this may correlate to the responsible person who recorded the values not having the necessary training making the missing values not MCAR but rather MAR or even worse

- Secondary source: recognize patterns in the missing values

Recognizing patterns in the missing values

- Analyse the percentage of missing values per attribute
  - Example: `df.isna().sum() / len(df) * 100`
- If there are very few missing values in a feature, look at the tuples
  - Example: `df[df.attr.isna()]`
- Use visualizations to find patterns
  - We will cover visualization libraries in depth in a later lecture!

Exercise

Exercise 2

Pima Indian Diabetes Dataset – MCAR, MAR or MNAR?

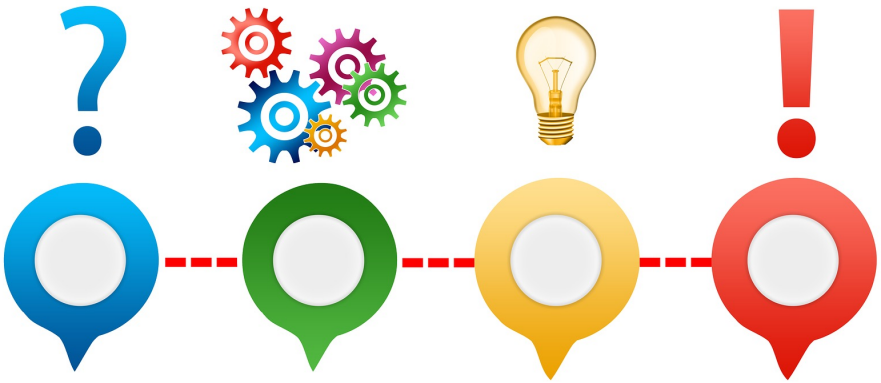


Photo by Gerd Altmann on Pixabay

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Handling Missing Data - Overview

The best solution to handle missing data is to have none!

can never be certain, if missing values are MCAR, MAR or MNAR due to *lurking variables*

Options for Handling Missing Data

Avoid Generating Missing Data

Just leave it as it is

Deletion

Simple Imputation

Forward/Backward Fill

Advanced Imputation

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Deletion

- Deleting Rows / Listwise Deletion / Complete Case Analysis
  - Delete all Rows (samples) with Missing Values
  - If missing values are rare and randomly distributed (MCAR)
  - If dropping rows does not significantly reduce the dataset
  - May introduce Bias, if MAR or MNAR
  - Example: `df_dropped_rows = df.dropna()`

returns all rows without na

- Deleting Columns
  - Delete all Columns with Missing Values (→ Drop Features)
  - If a column has a high percentage of missing values
  - If the missing column is not critical to analysis
  - May reduce predictive power of the model significantly (but does not introduce Bias)
  - Example: `df_dropped_cols = df.dropna(axis=1) or df.drop(columns=...)`

Simple Imputation: Replace missing values with Mean, Median or Mode

- Mean Imputation — mean more sensitive to outlier values
  - Works well for normally distributed numerical data (if MCAR)
  - Example: `df.attr = df.attr.fillna(df.attr.mean())`
- Median Imputation
  - Useful for skewed distributions (if MCAR)
  - Examples: `df.attr = df.attr.fillna(df.attr.median())`
- Mode Imputation — häufigster wert in der kategorie
  - Used for categorical data (if MCAR)
  - Examples: `df.attr = df.attr.fillna(df.attr.mode().iloc[0])`
- Common Mistake
  - When using train/test (or k-fold-cross validation), the imputed value must be computed on the training-set only (this makes train/test and esp. k-fold-cross-validation more complex to handle)
  - Better alternative: **`SimpleImputer()`**

Forward/Backward Fill

- Used for time-series data where missing values are sequential
- Forward Fill
  - Propagates previous values forward
  - Example: `df.Temperature = df.Temperature.fillna(method='ffill')`
- Backward Fill
  - Uses next valid observation to fill gaps
  - Example: `df.Temperature = df.Temperature.fillna(method='bfill')`
- Common Mistake
  - Make sure to not create train/test leakage, esp. during Backward Fill!

Advanced Imputation: K-NN Imputer

- K-Nearest Neighbors (K-NN) Imputation
  - Estimates missing values using the average of k-nearest neighbors
  - Works well when missing values have relationships with other variables
  - Need to define a suitable distance metric!
  - May give very good results
  - May introduce Bias
  - Computationally expensive
  - May be very hard to explain (“black box”)

  - Example:
```
knn_imputer = KNNImputer(n_neighbors=3)
df_knn = pd.DataFrame( knn_imputer.fit_transform(df),
                       columns=df.columns)
```

- Common Mistake
  - Predictor must be trained on the training-set only (or you may/will have “train-test leakage”)
  - Target Variable must not be used to impute features (or you may/will have “target leakage”)

Advanced Imputation: Regression Imputation

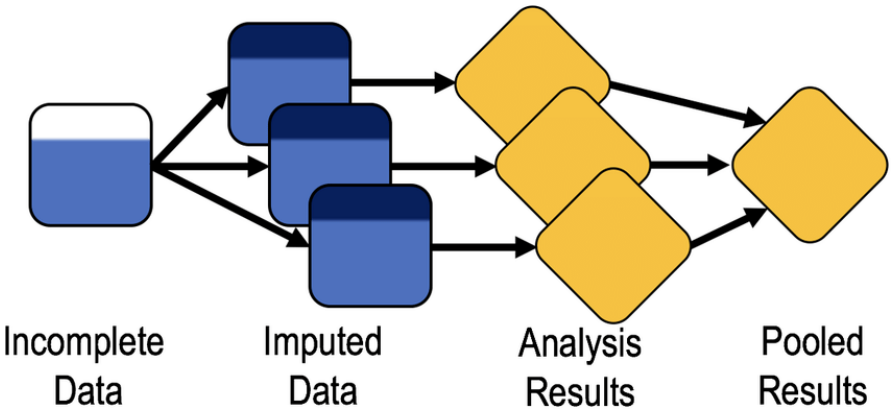
- Regression Imputation
  - Uses regression models to predict missing values based on available data
  - Best for datasets where missing values correlate with other features
  - May give very good results
  - May introduce Bias
  - Computationally expensive
  - May be hard to explain (“black box”)

- Common Mistake
  - Predictor must be trained on the training-set only (or you may/will have “train-test leakage”)
  - Target Variable must not be used to impute features (or you may/will have “target leakage”)

Advanced Imputation: Multiple Imputation

Multiple Imputation process

- missing data (shown in white) are imputed (shown in dark blue)
  ➔ M complete data sets (shown for M=3)
- Each complete imputed data set is analysed (e.g. classification, linear regression, …)
- Results are combined (Pooled)



- Open problem: how useful is single vs. multiple imputation in the context of prediction and classification when the user is not interested in measuring uncertainty due to missing values?

- Can be implemented in sklearn using the `IterativeImputer`, but requires additional work

statistically speaking would be correct to do this but too much work
interesting to execute on data set to see variance that is introduced via imputing (which may affect outcome result)

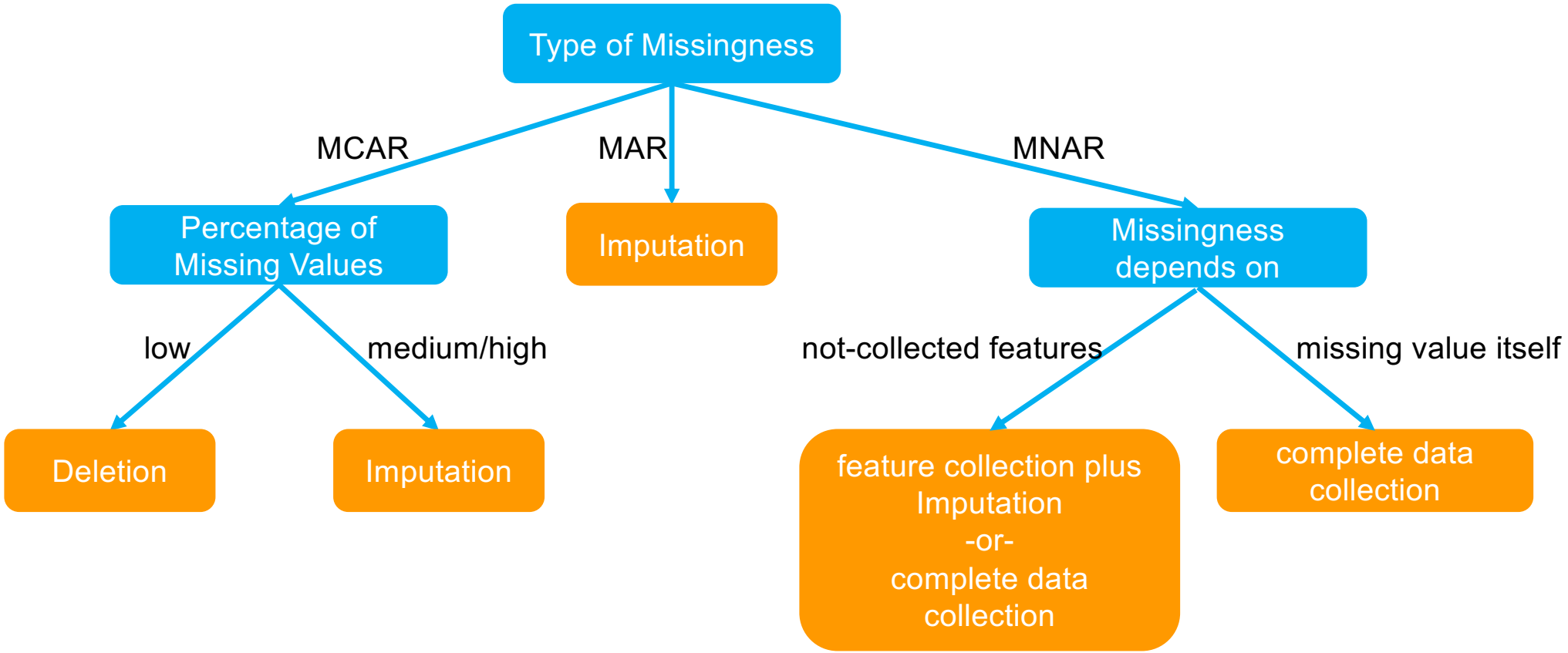
Source: https://www.researchgate.net/figure/The-multiple-imputation-MI-process-In-the-first-step-missing-data-shown-in-white-are_fig4_334213038

Comparing Different Missing Value Strategies

Some **Dropping** vs. **Imputation** Trade-offs

- Dropping reduces data size but avoids introducing noise
- Mean/median/mode imputation is simple but distorts distributions
- Forward/backward fill is good for time-series but can propagate incorrect values
- K-NN and regression imputation can work well for MAR but require assumptions

Missing Value Decision Tree



deletion introduces bias

Exercise

Exercise 3

Pima Indian Diabetes Dataset – Deletion and Imputation

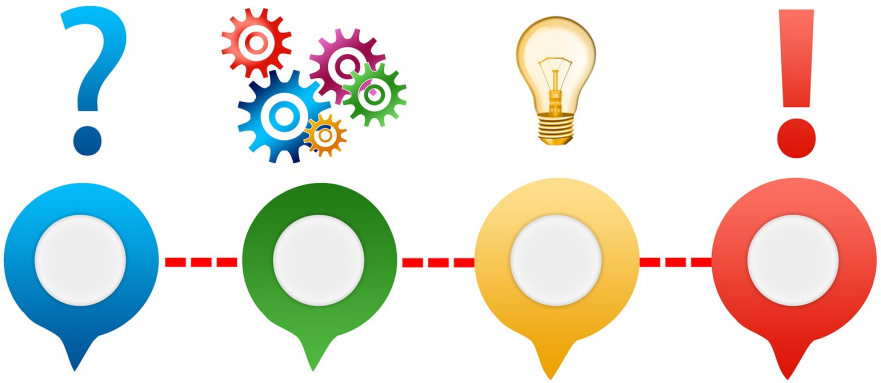
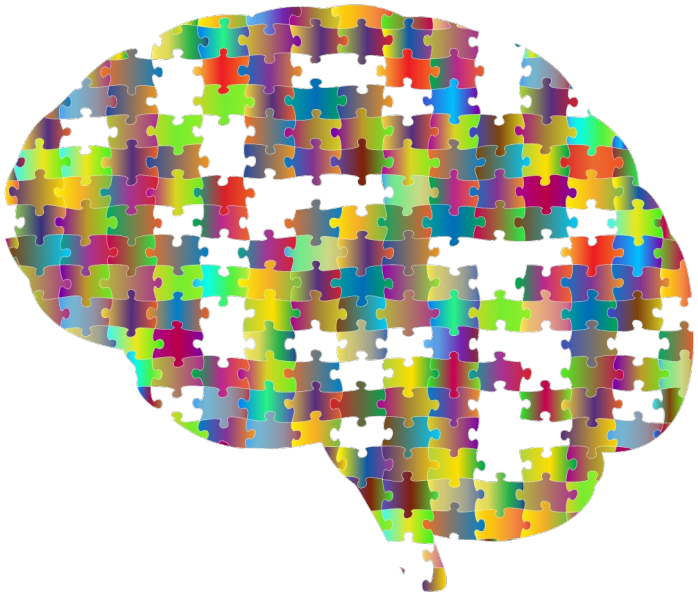


Photo by Gerd Altmann on Pixabay

Key Takeaways



- Missing values
  - very common in real-world dataset
  - can cause severe problems when not treated properly
- MCAR, MAR, MNAR: hard to decide, domain knowledge is key
- Deleting missing values
  - Only safe if MCAR
  - MAR, MNAR: will introduce bias in the model
  - Will reduce the power of the model
- (multiple) Imputation
  - best approach if MAR
- If MNAR has to be assumed, there is no good solution
  - If missingness depend on not-collected feature: collect these feature (so that MAR holds)
  - If missingness depends on the missing values themselves: collect data without missing values

Image by Gordon Johnson on pixabay